

High-Density High-Throughput WDM Optical Transmitter Based on Lithium-niobate-on-insulator Platform

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Abstract We propose an ultra-compact bragg microcavity modulator on LNOI platform, with a length of $\sim 100\mu\text{m}$, a measured 3-dB electro-optic bandwidth $> 110\text{GHz}$ and a modulation efficiency of $\sim 6.6\text{pm/V}$. It supports high-speed transmission of 140Gbit/s OOK and 180Gbit/s PAM4, enabling a 720Gbit/s four-channel transmitter. ©2026 The Author(s)

Introduction

Explosive global data traffic driven by artificial intelligence, large language models, 6G and autonomous driving is pushing data centers to their physical space and power limits, making high-speed, low-latency, high-density on-chip optical interconnects the key bottleneck in the compute-intensive era^[1-3]. Boosting the integration density of optical transceiver modules is thus critical to sustaining aggregate throughput growth within a constrained chip footprint^[4-6].

In high-capacity on-chip interconnect systems, the bandwidth and size of electro-optic modulators directly determine the system's speed limit and integration density. Although the thin-film lithium niobate (TFLN) platform offers significant advantages in high-speed modulation, traditional interferometric modulators typically require millimeter-scale lengths due to the small waveguide refractive index contrast, making high-density integration challenging^{[7], [8]}. While TFLN microring modulators can reduce device size to some extent, the small refractive index contrast necessitates larger bending radii, resulting in a typically small free spectral range (FSR) of approximately 1 nm, which is insufficient for wavelength division multiplexing applications^{[9], [10]}. Therefore, achieving deep miniaturization of devices while maintaining ultra-high-speed electro-optic response is crucial for advancing the TFLN platform towards large-scale integration.

In this work, we present a TFLN electro-optic modulator based on an ultra-short bragg cavity structure. The proposed design utilizes a Fabry-Perot (FP) cavity formed by multi-mode bragg gratings to achieve strong optical field localization, enabling high-efficiency electro-optic modulation within an extremely short device length, thereby significantly reducing the device size. With a device length of approximately $100\mu\text{m}$, the modulator achieves an electro-optic (EO)

bandwidth exceeding 110 GHz and a FSR of approximately 15 nm. Each channel supports 140 Gbit/s On-Off Keying (OOK) and 180 Gbit/s 4-level Pulse Amplitude Modulation (PAM4), resulting in a total throughput of 720 Gbit/s in a 4-channel Wavelength Division Multiplexing (WDM) system. This solution substantially reduces the footprint of the optical transmitter, significantly enhances the monolithic integration density of the Lithium Niobate on Insulator (LNOI) platform, and provides a core solution for next-generation high-density, ultra-high-speed WDM optical transmitters.

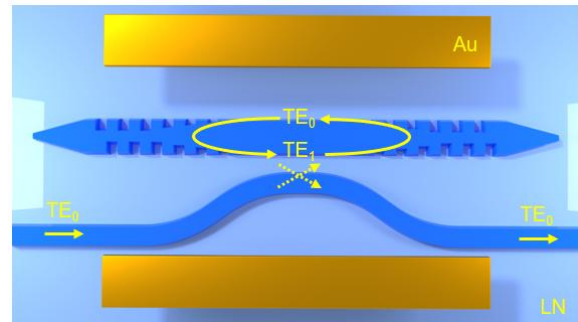


Fig. 1: Schematic configuration of the proposed ultra-compact and high-speed FP cavity EO modulator on an LNOI platform.

Device Design and Fabrication

Fig. 1 illustrates the structural schematic of the proposed bragg microcavity modulator. The device is designed on a LNOI platform, consisting of a pair of asymmetric multimode bragg grating (AMBG) reflectors and an access waveguide. The access waveguide serves to couple the input TE_0 mode into the FP cavity and convert it to the TE_1 mode. The AMBGs selectively reflect different modes, transforming the forward TE_1 (backward TE_0) mode into a backward TE_0 (forward TE_1) mode. Consequently, within the bragg reflection bandwidth, both forward TE_1 and backward TE_0 modes coexist in the FP cavity,

undergoing continuous conversion to form resonance. When the incident light meets the cavity's resonance condition, destructive interference occurs at the output port, resulting in strong suppression of light at the resonant wavelength, with a working mechanism analogous to the notch-type response of microring resonators. By adjusting the coupling gap G between the access waveguide and the cavity region, as well as the number of periods N in the bragg grating, the cavity's quality factor Q and extinction ratio can be flexibly controlled, enabling optimized trade-offs among modulation efficiency, bandwidth, and insertion loss. Ground-signal type electrodes are arranged on both sides of the device; when high-speed electrical signals are applied, a strong transverse electric field is generated in the lithium niobate waveguide. This field rapidly alters the effective refractive index through the Pockels effect of lithium niobate, causing dynamic shifts in the FP cavity's resonant wavelength and thereby achieving high-speed electro-optic modulation of the input optical signal.

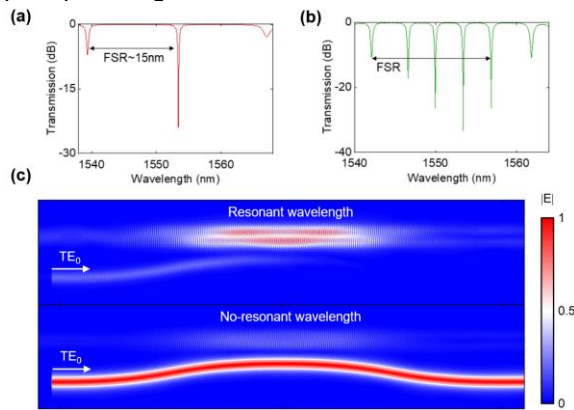


Fig. 2: (a) Calculated transmission spectrum of the FP cavity. (b) Simulated transmission spectra of four cascaded modulators with different cavity lengths. (c) Simulated light propagation profiles at resonant and non-resonant wavelengths.

The resonant operating wavelength of the device can be linearly tuned by precisely controlling the FP cavity length. Based on this principle, we demonstrate an ultra-compact on-chip multi-channel WDM optical transmitter. Fig. 2(a) and (b) show the simulated static transmission spectrum of the proposed FP cavity modulator, and the spectral characteristics of the four-channel optical transmitter array, respectively. Benefiting from the ultra-short cavity length of only 10 μm , the device achieves an ultra-large FSR of ~ 15 nm, which far exceeds that of conventional LNOI microring modulators, and fundamentally solves the core pain point that traditional microrings are incompatible with multi-channel WDM due to insufficient FSR.

Such an ultra-large FSR provides a sufficient spectral window for on-chip multi-channel multiplexing on the LNOI platform. On this basis, we

designed and fabricated a four-channel ultra-compact WDM optical transmitter with a uniform channel spacing of 3.2 nm, which is realized by a 100 nm cavity length increment between adjacent channels. The simulation results show that the resonant peaks of the four channels are uniformly distributed in the wavelength range from 1538 nm to 1553 nm, with no obvious inter-channel crosstalk. All channels maintain consistent extinction ratio and resonance characteristics, exhibiting excellent array uniformity. The simulated optical field propagation results in Fig. 2 (c) further verify the operating mechanism of the device: at the resonant wavelength, the optical field is effectively confined in the FP cavity to achieve high-efficiency electro-optic modulation, while the optical signal passes through the waveguide with low loss at non-resonant wavelengths, which lays a solid foundation for multi-channel parallel modulation.

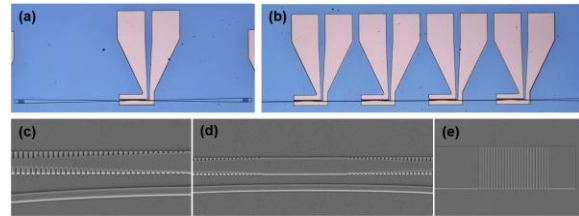


Fig. 3: (a) Fabricated FP-cavity modulator and (b) the cascaded modulators designed for DWDM. Scanning electron microscope (SEM) images of the (c) AMBG, (d) the coupling region, and (e) the TE grating coupler.

The devices are fabricated via a fully CMOS-compatible process. Waveguide and grating structures are patterned by 100 kV electron-beam lithography (EBL), followed by low-damage Ar^+ plasma etching with a precisely controlled 150 nm etch depth. After depositing a 1 μm -thick SiO_2 upper cladding via plasma-enhanced chemical vapor deposition, Ti/Au (20 nm/1.2 μm) coplanar driving electrodes are formed via electron-beam evaporation and a standard lift-off process. Fig. 3 presents the metallurgical microscopy images and scanning electron microscope (SEM) characterization results of the fabricated FP cavity electro-optic modulator

Characterization

Static spectral tests are performed with a C-band tunable laser and high-precision optical power meter, with all results normalized to a reference straight waveguide on the same chip to eliminate coupling and fiber link loss. As shown in Fig. 4(a), (b) and (c), The fabricated single-channel FP modulator exhibits a low excess loss of ~ 0.2 dB, a high on-off ER of ~ 24 dB, a loaded Q-factor of ~ 2600 , and an ultra-large FSR of ~ 15 nm, which is in good agreement with simulation results. The 4-channel transmitter shows four uniform resonant dips in the 1538–1553 nm wavelength range, with a consistent 3.2 nm channel spacing, an

average excess loss of ~ 0.8 dB, and a minimum ER > 20 dB across all channels.

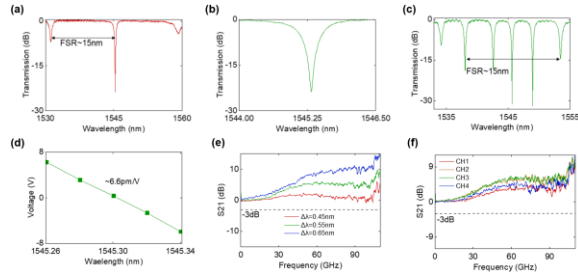


Fig. 4: (a) Measured transmission spectrum of the presented FP-cavity modulator. (b) Enlarged view at the resonant wavelength. (c) Measured optical spectral response of the four-channel transmitter. (d) Resonant wavelengths corresponding to different applied voltages. (e) measured small-signal EO response (S21) of the single modulator under different wavelength detunings ($\Delta\lambda$). (f) EO responses of the four-channel transmitter.

Fig. 4(d) depicts the measured results of low-frequency electro-optic characterization, showing a tuning efficiency of ~ 6.6 pm/V, which can be further improved to ~ 11 pm/V by reducing the electrode gap to 3 μm . The EO bandwidth are also recorded in Fig. 4(e) and (f), which carried out with a lightwave component analyzer (LCA). The modulator achieves a 3-dB EO bandwidth exceeding 110 GHz with a 0.45 nm wavelength detuning. All 4 channels of the transmitter exhibit consistent > 110 GHz 3-dB EO bandwidth, with less than 5 GHz variation across channels, demonstrating excellent uniformity for parallel high-speed operation.

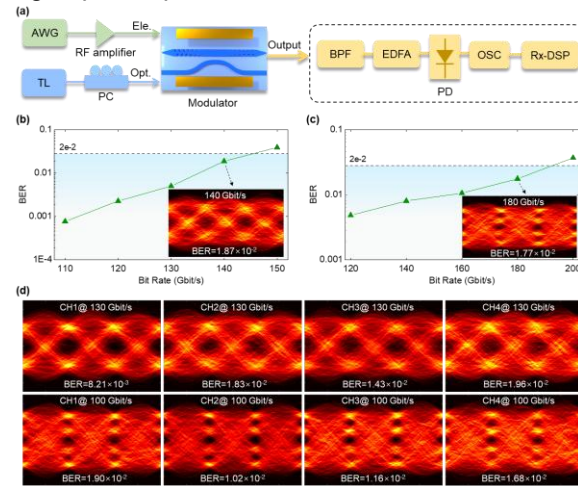


Fig. 5: (a) High-speed data transmission measurement setup. Measured BER performance of (b) OOK and (c) PAM4 as a function of bit rate. The insets display the eye diagrams for 140 Gbit/s OOK and 180 Gbit/s PAM4. (d) Measured eye diagrams of 140 Gbit/s OOK and 180 Gbit/s PAM4 for four-channel transmitter.

Back-to-back high-speed data transmission experiments are conducted with a 70 GHz arbitrary waveform generator, 55 GHz broadband electrical amplifier, and offline digital signal processing (DSP). The single modulator achieves error-free transmission below the soft-decision forward error correction (SD-FEC) threshold of

2×10^{-2} at 140 Gbit/s OOK and 180 Gbit/s PAM4, with clear and open eye diagrams for both formats. For the 4-channel WDM transmitter, all channels support error-free 180 Gbit/s PAM4 transmission, delivering an aggregate throughput of 720 Gbit/s. The modulator achieves a power dissipation per bit (PDP) as low as 13.9 fJ/bit at 180 Gbit/s PAM4, an order of magnitude lower than conventional LNOI microring modulators, showing significant low-power advantages for high-density optical interconnects.

Conclusions

We experimentally demonstrate an ultra-compact 4-channel WDM optical transmitter on the LNOI platform. The proposed AMBG-based FP cavity modulator eliminates extra mode multiplexers, achieving a footprint of only 100 μm with a $\sim 5\times$ size reduction over prior arts. The device exhibits > 110 GHz electro-optic bandwidth, ~ 15 nm FSR, and ~ 24 dB extinction ratio, supporting 180 Gbit/s PAM4 transmission per channel with an aggregate throughput of 720 Gbit/s. With a record-low PDP of 13.9 fJ/bit and > 1 Tbit/s scalability, this work provides a key solution for high-density ultra-high-speed LNOI optical transmitters in next-generation optical interconnects.

Acknowledgements

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